

Activity 8.25

Using pseudocode write an algorithm to copy a line of text from one text file to another text file.

In this chapter, you have learnt about:

- ✓ declare, use and identify appropriate data types for variables and constants
- ✓ understand and use input and outputs
- ✓ understand and use the programming concepts: sequence, selection, iteration, totalling, counting and string handling
- ✓ understand and use nested statements
- ✓ understand and use arithmetic, logical and Boolean operators
- ✓ understand and use procedures, functions and library routines
- ✓ write maintainable programs
- ✓ declare, use, identify appropriate data types for, read and write arrays with one and two dimensions
- ✓ store data in a file and retrieve data from a file.

Key terms used throughout this chapter

variable – a named data store that contains a value that may change during the execution of a program

constant – a named data store that contains a value that does not change during the execution of a program

declare – define the value and data type of a variable or constant

integer – a positive or negative whole number that can be used with mathematical operators

real number – a positive or negative number with a fractional part; Real numbers can be used with mathematical operators

char – a variable or constant that is a single character

string – a variable or constant that is several characters in length. Strings vary in length and may even have no characters (an empty string); the characters can be letters and/or digits and/or any other printable symbol

sequence – the order in which the steps in a program are executed

selection – allowing the selection of different paths through the steps of a program

iteration – a section of programming code that can be repeated under certain conditions

counting – keeping track of the number of times an action is performed

totalling – keeping a total that values are added to

operator – a special character or word in a programming language that identifies an action to be performed

arithmetic operator – an operator that is used to perform calculations

logical operator – an operator that is used to decide the path to take through a program if the expression formed is true or false

Boolean operator – an operator that is used with logical operators to form more complex expressions

nesting – the inclusion of one type of code construct inside another

procedure – a set of programming statements grouped together under a single name that can be called to perform a task in a program, rather than including a copy of the code every time the task is performed

function – a set of programming statements grouped together under a single name which can be called to perform a task in a program, rather than including a copy of the code every time; just like a procedure except a function will return a value back to the main program

parameters – the variables in a procedure or function declaration that store the values of the arguments passed from the main program to a procedure or function

MOD – an arithmetic operator that returns the remainder of a division; different languages use different symbols for this operation

DIV – an arithmetic operator that returns the quotient (whole number part) of a division; different languages use different symbols for this operation

ROUND – a library routine that rounds a value to a given number of decimal places

RANDOM – a library routine that generates a random number

array – a data structure containing several elements of the same data type; these elements can be accessed using the same identifier name

index – identifies the position of an element in an array

file – a collection of data stored by a computer program to be used again